

# IPCC Tier 2 Livestock GHG Uncertainty Calculator

Pre-Flight Questionnaire — GMH Uncertainty Translator

CIAT / CGIAR Alliance — ClimateActionNetZero

Fill this out **before** opening the Translator on claude.ai. When you start the chat, paste the contents of this filled page as your first message — Claude will use it to skip the warm-up questions and go straight to inspecting your data.

*Save this file as `my_inventory_questionnaire.docx` and type your answers directly into the form below. Italic lines are guidance and can be left in place.*

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## 1. Country and inventory year

- **Country:** \_\_\_\_\_
- **Continental region** (tick one — drives the IPCC BW deviation benchmark): ☐ africa ☐ asia ☐ europe ☐ americas ☐ oceania ☐ global
- **Inventory year being prepared:** \_\_\_\_\_

## 2. IPCC guidelines version

Tick one:

- ☐ **IPCC 2006 Guidelines** (still the default reporting requirement in many countries)
- ☐ **IPCC 2019 Refinement** (newer; updated emission factors, more manure systems)

*If unsure, ask your national focal point or pick 2006 — the app accepts both.*

## 3. Cattle sub-categories

*List the sub-categories your inventory splits cattle into and the approximate head count of each. You can use any names — Claude will map them to the app's controlled vocabulary.*

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| Sub-category name | Head count | Sex (M/F/mix) | Age class (adult / 1–3yr / calf) |
|-------------------|------------|---------------|----------------------------------|
|-------------------|------------|---------------|----------------------------------|

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*Example row: dairy cows / 500,000 / F / adult*

## 4. Manure management

*For each sub-category, roughly what % of manure goes to each system? Per sub-category, the percentages must sum to 100. Recognised systems:*

- pasture, daily\_spread, solid\_storage, solid\_storage\_covered\*, dry\_lot, deep\_bedding
- liquid\_slurry, lagoon, composting, anaerobic\_digester\*, aerobic\_treatment\*, burned\_for\_fuel\*

*\*available only when IPCC version = 2019 Refinement.*

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| Sub-category | System 1 (%) | System 2 (%) | System 3 (%) | System 4 (%) |
|--------------|--------------|--------------|--------------|--------------|
|--------------|--------------|--------------|--------------|--------------|

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*Example: dairy cows: pasture 40%, solid\_storage 35%, liquid\_slurry 15%, lagoon 10% (sums to 100).*

**Optional — is the allocation itself uncertain?** If yes, give a range per share (e.g. “pasture 40 %, lower 30 %, upper 55 %”). The app samples each MMS within its range and renormalises every Monte Carlo iteration so the totals still sum to 100 %. Leave blank if you are confident in the central allocation.

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| Sub-category | System | Central % | Lower % | Upper % |
|--------------|--------|-----------|---------|---------|
|--------------|--------|-----------|---------|---------|

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## 5. Data fields you have

*Tick the parameters for which you have country-specific values. Unticked parameters will be filled with IPCC defaults and flagged amber in the app's QA/QC tab for review.*

### Activity data (the population number)

- ☐ Animal population (head count) — **strongly recommended**

### Animal characteristics

- ☐ Body weight (kg) — **BW**  
☐ Mature body weight (kg) — **MW**  
☐ Daily weight gain (kg/day) — **WG**  
☐ Fraction of females pregnant in the year (includes pregnant heifers) — **pct\_pregnant**

### Production (dairy only)

- ☐ Daily milk yield per lactating cow (kg/day) — **Milk**  
☐ Milk fat content (%) — **Fat**  
☐ Milk protein content (%) — **MilkPR**

### Feed

- ☐ Digestible energy of feed (% of gross energy) — **DE**  
☐ Crude protein in feed (%) — **CP**

### Climate (optional)

- ☐ Mean winter temperature (°C) — **Tw**

**IPCC equation coefficients** (*rarely measured locally; leave unticked unless your country has a published value*)

- ☐ **C<sub>f</sub>i, C<sub>a</sub>, C, C<sub>p</sub>, Y<sub>m</sub>, B<sub>o</sub>, A<sub>SH</sub>, U<sub>E</sub>**  
☐ Pasture/indirect N<sub>2</sub>O emission factors (EF3\_PRP, EF4, EF5, Frac\_GASM\_PRP, Frac\_LEACH\_PRP)

*Managed-storage manure-N<sub>2</sub>O values (the direct EF and the volatilisation / leaching fractions) are entered **per manure-management system** in the Manure Management section above, not as single Parameters values.*

## 6. Where do your uncertainty estimates come from?

Tick one (or combine if different sources for different parameters):

- ☐ (a) **No uncertainty estimates** — use IPCC suggested  $\pm\%$  from the catalogue  
☐ (b) **Expert judgement  $\pm\%$**  — I'll write a single  $\pm\%$  next to each value  
☐ (c) **Measured confidence intervals** — I have explicit lower / upper bounds (or  $\pm$  with

sample size)

- ☐ (d) **A mix** — I'll specify per parameter

## 7. Time series (optional)

- ☐ I have annual data for **5 or more years** that the app can use to estimate correlations between parameters.

• **Earliest year:** \_\_\_\_\_ **Latest year:** \_\_\_\_\_

• **Parameters covered (tick all that apply):**

- ☐ N
- ☐ BW
- ☐ MW
- ☐ WG
- ☐ Milk
- ☐ Fat
- ☐ pct\_pregnant
- ☐ DE
- ☐ CP
- ☐ MilkPR

## 8. File(s) you'll upload

*List the file names and a one-line description of what's in each.*

1. \_\_\_\_\_
2. \_\_\_\_\_
3. \_\_\_\_\_

## 9. Anything else you want Claude to know

*Free-form note: anything unusual about your data, sub-category definitions specific to your country, or known quirks. Optional.*

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**That's it.** Paste this filled page into the Translator and attach your data files. Claude will take it from here.